

Algorithm and Flowchart

1. A university cafeteria offers discounts to students based on the following rules:

1. If a student is a member of the Cafeteria Club, they get a **20% discount** on their total bill.
2. If a student is not a club member but their total bill is more than **500 units (currency)**, they get a **10% discount**.
3. If neither condition is met, no discount is applied.

Write an algorithm to calculate the final amount a student needs to pay based on the total bill and whether they are a club member also draw flowchart

2. A university wants to automate the grading system for a course based on the following rules:

1. If a student's score is **90 or above**, they receive a grade of **A**.
2. If the score is **80 to 89**, the grade is **B**.
3. If the score is **70 to 79**, the grade is **C**.
4. If the score is **60 to 69**, the grade is **D**.
5. If the score is below **60**, the grade is **F**.

Write an algorithm to determine a student's grade based on their score and also draw the flowchart.

3. An amusement park has specific rules for allowing visitors on a new roller coaster ride. The rules are as follows:

1. A visitor must be at least **12 years old** to ride.
2. The visitor's height must be **at least 140 cm**.
3. If the visitor is under **18 years old**, they must have a parent or guardian's permission.

Write an algorithm to determine if a visitor is eligible to ride the roller coaster and also draw the flowchart.

4. A bank wants to automate the process of checking loan eligibility based on the following rules:

1. The applicant must be at least **21 years old**.
2. The applicant's monthly income must be **at least 30,000 units (currency)**.
3. If the applicant's income is between **30,000 and 50,000**, they can only apply for a loan amount of up to **500,000 units**.
4. If the income is above **50,000**, they can apply for a loan amount of up to **1,000,000 units**.
5. Applicants who are under 21 years old or have a monthly income below 30,000 are **not eligible** for a loan.

Write an algorithm to determine if an applicant is eligible for a loan and the maximum loan amount they can apply for.

5. An online store provides discounts to customers based on the total purchase amount and membership status:

1. If a customer is a **Premium Member**, they get a **20% discount** on any purchase.
2. If a customer is not a member but their total purchase amount exceeds **500 units (currency)**, they get a **10% discount**.
3. If the purchase amount is **1,000 units or more**, all customers (both members and non-members) get an **additional 5% discount** on the total after applying the first discount.
4. If neither condition applies, no discount is given.

Write an algorithm to calculate the final amount a customer needs to pay based on their membership status and purchase amount.

6. A company calculates employee bonuses based on the following criteria:

1. If an employee's **performance rating** is "**Excellent**", they receive a **20% bonus** of their salary.
2. If the performance rating is "**Good**", they receive a **10% bonus** of their salary.
3. If the performance rating is "**Average**", they receive a **5% bonus** of their salary.
4. If the performance rating is "**Poor**", no bonus is given.
5. In addition, if an employee has worked for **5 or more years**, they will receive an **extra 5% bonus** on top of their performance-based bonus.

Write an algorithm to calculate the bonus an employee should receive based on their performance rating and years of service.

7. A library charges fines for late returns of books based on the following rules:

1. If the book is returned within **5 days after the due date**, the fine is **2 units (currency) per day**.
2. If the book is returned between **6 to 10 days** late, the fine is **5 units per day**.
3. If the book is returned more than **10 days** late, the fine is **10 units per day**.
4. If the book is returned **on or before the due date**, there is **no fine**.
5. The maximum fine is capped at **500 units**.

Write an algorithm to calculate the fine for a late return based on the number of days late.

8. An electricity provider calculates monthly bills for residential customers based on the following rates:

1. For the first **100 units**, the charge is **5 units (currency) per unit**.
2. For the next **101 to 300 units**, the charge is **7 units per unit**.
3. For any usage above **300 units**, the charge is **10 units per unit**.
4. A fixed **service charge of 50 units** is added to all bills.

Write an algorithm to calculate the total electricity bill based on the number of units consumed.

9. A movie theater calculates ticket prices based on the customer's age and the show timing:

1. **Age-based discounts:**

- Children (age 0–12): **50% discount** on the standard price.
- Seniors (age 60+): **30% discount** on the standard price.
- Regular (age 13–59): **No discount** on the standard price.

2. **Show timing discounts:**

- Morning shows (before 12 PM): **20% discount** on the ticket price (after applying any age-based discounts).
- Evening shows (after 6 PM): **No additional discount**.

3. The standard ticket price is **300 units**.

Write an algorithm using to calculate the final ticket price for a customer based on their age and the show timing.